

MENDIG, Germany

WMO: 105140

Lat: 50.366N

Lon: 7.315E

Elev: 597

StdP: 14.38

Time Zone: 1.00 (EUC)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	19.3	11.5	10.1	17.0	14.7	11.8	22.4	28.3	46.1	25.0	45.5	3.5	240	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.0	88.8	67.6	85.3	66.5	81.6	65.2	69.8	84.2	68.2	81.9	66.4	78.6	7.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.9	94.2	74.0	63.3	89.0	72.8	61.7	84.2	71.5	34.0	84.6	32.7	81.7	31.4	78.6	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.0	15.6	DB	8.0	95.2	5.9	3.4	3.8	97.6	0.3	99.6	-3.0	101.5	-7.2	103.9	(4)
(5)			WB	7.8	73.4	5.8	1.6	3.6	74.5	0.2	75.4	-3.1	76.3	-7.3	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	34.9	38.0	43.4	48.9	56.5	62.3	66.0	66.8	58.5	50.5	42.1	36.0	(6)
(7)		DBStd	12.91	8.28	7.47	6.50	6.44	6.30	6.28	5.87	6.00	5.07	6.20	6.12	7.78	(7)
(8)		HDD50	1893	469	339	222	98	13	0	0	0	2	70	244	436	(8)
(9)		HDD65	5596	933	757	670	483	272	128	60	50	206	451	688	898	(9)
(10)		CDD50	2031	1	2	16	64	215	369	495	520	256	85	6	2	(10)
(11)		CDD65	262	0	0	0	0	9	47	90	105	10	1	0	0	(11)
(12)		CDH74	3170	0	0	0	11	195	581	1020	1221	138	4	0	0	(12)
(13)		CDH80	1092	0	0	0	0	37	171	370	483	31	0	0	0	(13)
(14)	Wind	WSAvg	6.3	7.4	7.7	7.0	6.6	5.8	5.8	5.9	5.7	6.0	6.2	5.8	6.6	(14)
(15)	Precipitation	PrecAvg	26.9	2.0	1.9	1.9	1.7	2.4	2.7	2.8	2.5	2.4	2.1	2.2	2.4	(15)
(16)		PrecMax	33.5	4.9	4.3	4.1	3.4	5.2	5.6	6.5	5.4	5.8	5.0	4.1	5.7	(16)
(17)		PrecMin	21.3	0.4	0.4	0.4	0.0	0.5	0.9	1.1	0.4	0.9	0.6	0.2	0.7	(17)
(18)		PrecStd	3.3	0.9	0.9	0.9	0.8	1.1	1.1	1.2	1.3	1.3	1.0	0.9	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	57.5	67.7	75.6	84.6	89.8	92.6	95.3	84.8	73.0	58.9	54.8	(19)
MCWB			50.0	49.2	54.2	58.9	67.2	68.1	69.1	69.6	64.4	63.6	52.9	50.9	(20)	
2%		DB	51.7	53.9	61.8	69.9	79.1	84.8	88.3	89.6	77.9	66.9	55.3	52.5	(21)	
		MCWB	47.8	47.8	51.6	56.0	63.3	67.2	67.5	68.0	63.4	58.9	50.9	49.2	(22)	
5%		DB	48.7	51.4	57.6	65.6	75.1	80.9	84.2	85.6	73.2	63.4	53.1	50.1	(23)	
		MCWB	45.5	46.7	49.6	53.9	60.9	65.2	66.6	66.1	60.9	57.6	49.4	47.4	(24)	
10%		DB	46.0	49.0	54.1	61.7	70.1	76.6	79.9	81.2	69.0	60.1	51.0	47.3	(25)	
		MCWB	43.2	45.0	48.3	51.9	58.5	63.4	64.7	65.1	59.0	55.1	48.1	45.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	50.9	55.8	61.3	68.3	71.5	71.8	71.7	67.5	64.3	54.2	51.6	(27)
MCD B			54.0	55.5	64.0	72.0	80.7	84.6	87.8	87.3	79.3	71.9	56.8	53.8	(28)	
2%		WB	47.9	48.8	52.9	57.3	64.6	68.8	69.3	69.6	64.5	60.3	51.8	49.5	(29)	
		MCD B	51.0	53.3	59.8	66.9	78.2	81.4	83.7	85.1	75.3	65.7	54.6	52.1	(30)	
5%		WB	45.6	47.2	50.9	54.9	61.7	66.5	67.7	68.0	62.3	57.9	49.9	47.4	(31)	
		MCD B	48.5	50.9	56.1	63.5	72.7	79.2	81.1	82.1	71.0	62.4	52.5	49.8	(32)	
10%		WB	43.2	45.3	48.8	53.0	59.3	64.1	65.8	66.1	60.1	55.5	48.1	44.9	(33)	
		MCD B	45.7	48.8	53.6	60.5	68.6	74.5	77.2	77.9	67.2	59.6	50.8	47.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.5	12.8	16.0	19.4	20.4	21.7	21.1	22.0	18.6	15.9	11.1	9.7	(35)
MCDBR			12.4	15.8	22.9	27.2	29.5	30.0	29.9	30.6	27.1	20.4	14.0	10.6	(36)	
MCWBR			9.9	10.8	14.2	14.9	15.6	14.4	12.7	12.5	13.8	13.3	10.4	8.7	(37)	
5% WB		MCDBR	12.2	14.2	19.4	24.2	27.2	26.4	26.2	25.9	23.2	19.1	13.1	10.2	(38)	
		MCWBR	10.1	10.4	12.5	13.7	14.9	13.7	11.9	11.5	12.5	12.9	10.4	8.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.341	0.375	0.405	0.411	0.414	0.415	0.410	0.381	0.364	0.338	0.313	(40)	
taud		2.413	2.353	2.280	2.240	2.256	2.269	2.259	2.303	2.363	2.399	2.406	2.401	(41)		
Ebn at Noon		228	250	260	265	268	268	266	261	256	238	216	210	(42)		
Edh at Noon		20	27	34	40	41	41	41	37	31	26	20	17	(43)		
(44)	All-Sky Solar Radiation	RadAvg	243	464	825	1312	1542	1703	1611	1384	1029	598	277	186	(44)	
(45)		RadStd	38	88	126	156	148	161	186	137	128	61	36	27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page